Generalized Sum Rank Metric for Convolutional Codes

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Abstract

In this paper, we introduce the generalized sum rank metric for convolutional codes, which extends the Hamming metric and the (sum) rank metric for both block and convolutional codes. This new metric allows the information sent at each time instant to be divided into smaller packets of arbitrary and different sizes. We derive a generalized Singleton bound for codes in this metric and present constructions that achieve this bound. In addition, we study column distances in the generalized sum rank metric, provide upper bounds, and give optimal code constructions.

Key Words : Convolutional codes, Sum-rank metric, Construction of optimal codes

1 Introduction

When using error-correcting codes for application in random linear network coding, the suitable framework is to consider codes endowed with the rank metric instead of the classical Hamming metric. For the case of multi-shot network coding, i.e. when a network is used several times to transmit various packages of information, the proposed measure for the involved error-correcting codes is the sum rank metric (see e.g. [NUF10, SK20]), which turns out to be a generalization of both, Hamming metric and rank metric for linear block codes; for more details on codes in the sum-rank metric see e.g. [GMPS23, BGLR21]. In this setting, codes are used in

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several time instances, where dependencies between the packages sent in the different shots should be created and where the overall metric can be written as the sum of the metrics for the single time instances. This naturally suggests the use of convolutional codes for multishot network coding and hence, rank metric convolutional codes have been studied in e.g. [WZSS15, AMPN20, NPS17, MBK16]. When using convolutional codes in the rank metric, the analogue to the free distance, i.e. the classical minimum distance in the Hamming metric, is already called sum rank distance as the structure of convolutional codes inherently connects the encoding of different time instants and calculating distances in general involves forming a sum of distances corresponding to different time instants. The same is true if one considers column distances in the rank metric (see e.g. [MBK16, NPRS17]), which are the crucial measure for low-delay applications.

In this paper, we propose an even more general point of view by introducing the generalized sum rank distance for convolutional codes, which is a generalization of the Hamming metric and the (sum) rank metric for convolutional codes as well as a generalization of the sum rank metric for block codes. The use of codes in this new metric for multishot network coding allows to split the amount of information sent at one time instant into smaller packages of arbitrary and different sizes. We show a generalized Singleton bound for convolutional codes in the generalized sum rank metric and present constructions for codes that reach this bound. For these constructions we make use of rank metric convolutional codes that reach the Singleton bound in the usual (sum) rank metric. Moreover, we consider column distances in the generalized sum rank metric, show upper bounds for these distances and provide constructions for convolutional codes that are optimal with respect to these bounds. Furthermore, we improve upon existing constructions for rank metric convolutional codes via showing how to obtain rank metric convolutional codes with optimal column distances from certain MRD block codes. In [MPN20, Definition 12] a generalized sum rank metric (that is just called sum rank metric there) has already been defined for vector rank metric convolutional codes. However, in [MPN20] these codes are not studied in detail but are mainly used for the construction of so-called partial MDP codes. Also in this paper, we take a different point of view by considering matrix convolutional codes in the generalized sum rank metric, which leads to upper bounds for sum rank column distances that are much more involved than and differ from the bounds presented in [MPN20, Proposition 4].

2 Preliminaries

The following definition generalizes Hamming metric and rank metric for block codes. To this end we set $F := \mathbb{F}_q^{n_1 \times m_1} \times \cdots \times \mathbb{F}_q^{n_t \times m_t}$.

Definition 1. *Let* $t, m_j, n_j \in \mathbb{N}$*,* $j = 1, \ldots, t$*,* $\psi : \mathbb{F}_q^{n_1m_1+\cdots+n_tm_t} \to F$*,* $v = (v_1 \cdots v_t) \mapsto (V_1, \ldots, V_t)$*,* $v_j = (v_{1,1}^{(j)}, \ldots, v_{1,m_j}^{(j)}, \ldots, v_{n_j,1}^{(j)}, \ldots, v_{n_j,m_j}^{(j)}) \in \mathbb{F}_q^{n_jm_j}$*,* $V_j = (v_{i,\ell}^{(j)}) \in \mathbb{F}_q^{n_j\times m_j}$*. A **sum rank metric block code** $\mathcal{C}$ is the image of a monomorphism* $\varphi = \psi \circ \gamma$ *for some monomorphism* $\gamma : \mathbb{F}_q^k \to \mathbb{F}_q^{n_1m_1+\cdots+n_tm_t}$*, i.e. there exists a matrix* $G \in \mathbb{F}_q^{k\times(n_1m_1+\cdots+n_tm_t)}$*, called a **generator matrix** of* $\mathcal{C}$*, such that* $\gamma(u) = uG = v$*. For* $A = (A_1, \ldots, A_t), B = (B_1, \ldots, B_t) \in F$*, the **sum rank distance** is defined as*

$$d_{SR}(A, B) = wt_{SR}(A - B) = \sum_{j=1}^{t} wt_R(A_j - B_j) = \sum_{j=1}^{t} rank(A_j - B_j)$$

*and the **(minimum) sum rank distance** of* $\mathcal{C}$ *is*

$$d_{SR}(\mathcal{C}) = \min_{A,B\in\mathcal{C},A\neq B} d_{SR}(A, B) = \min_{C\in\mathcal{C}\setminus\{0\}} wt_{SR}(C).$$

The following theorem generalizes the Singleton bound for Hamming metric and rank metric block codes.

Theorem 1 ([BGLR21])**.** *Let* $\mathcal{C} \subseteq \mathbb{F}_q^{n_1\times m_1} \times \cdots \times \mathbb{F}_q^{n_t\times m_t}$ *be a sum rank metric code with* $|\mathcal{C}| \geq 2$ *and* $d_{SR}(\mathcal{C}) = d$*. Let* ℓ *be the unique integer satisfying* $0 \leq d - 1 - \sum_{i=\ell+1}^{t} n_i \leq n_\ell - 1$*. Then,* $dim(\mathcal{C}) \leq \sum_{i=1}^{\ell} m_in_i - m_\ell(d - 1 - \sum_{i=\ell+1}^{t} n_i)$*. Codes reaching equality in this bound are called **Maximum Sum Rank Distance (MSRD)** block codes. For* $t = 1$ *such codes are called **Maximum Rank Distance (MRD)**.*

Next, we generalize Definition 1 to the setting of convolutional codes. In this way, we also get a generalization of both Hamming metric and rank-metric convolutional codes. We define $R := \mathbb{F}_q[D]^{n_1\times m_1} \times \cdots \times \mathbb{F}_q[D]^{n_t\times m_t}$.

Definition 2. *Let* $t, m_j, n_j \in \mathbb{N}$*,* $j = 1, \ldots, t$*,* $\psi : \mathbb{F}_q[D]^{n_1m_1+\cdots+n_tm_t} \to R$*,* $v(D) = (v_1(D) \cdots v_t(D)) \mapsto (V_1(D), \ldots, V_t(D))$*,* $v_j(D) = (v_{1,1}^{(j)}(D), \ldots, v_{1,m_j}^{(j)}(D), \ldots, v_{n_j,1}^{(j)}(D), \ldots, v_{n_j,m_j}^{(j)}(D)) \in \mathbb{F}_q[D]^{n_jm_j}$*,* $V_j(D) = (v_{i,\ell}^{(j)}(D)) \in \mathbb{F}_q[D]^{n_j\times m_j}$*. A **sum rank metric convolutional code** $\mathcal{C}$ is the image of a monomorphism* $\varphi = \psi \circ \gamma$ *for some monomorphism* $\gamma : \mathbb{F}_q^k[D] \to \mathbb{F}_q[D]^{n_1m_1+\cdots+n_tm_t}$*, i.e. there is a matrix* $G(D) \in \mathbb{F}_q[D]^{k\times(n_1m_1+\cdots+n_tm_t)}$ *with full rank, called a **generator matrix** of* $\mathcal{C}$*, such that* $\gamma(u(D)) = u(D)G(D) = v(D)$*. For*

$$A(D) = (A_1(D), \ldots, A_t(D)) = (\sum_{i\in\mathbb{N}} A_{1,i}D^i, \ldots, \sum_{i\in\mathbb{N}} A_{t,i}D^i) \in R$$

$$B(D) = (B_1(D), \ldots, B_t(D)) = (\sum_{i\in\mathbb{N}} B_{1,i}D^i, \ldots, \sum_{i\in\mathbb{N}} B_{t,i}D^i) \in R$$

*the **generalized sum rank distance** is defined as*

$$\begin{aligned} d_{SR}(A(D),B(D)) &= wt_{SR}(A(D)-B(D)) \\ &= \sum_{j=1}^{t} wt_R(A_j(D)-B_j(D)) = \sum_{j=1}^{t}\sum_{i\in\mathbb{N}} rank(A_{j,i}-B_{j,i}) \end{aligned}$$

*and the **(minimum) generalized sum rank distance** of $\mathcal{C}$ is*

$$d_{SR}(\mathcal{C}) = \min_{\substack{A(D),B(D)\in\mathcal{C},\\ A(D)\neq B(D)}} d_{SR}(A(D),B(D)) =: \min_{C(D)\in\mathcal{C}\setminus\{0\}} wt_{SR}(C(D)).$$

If we set $t=1$ in the previous definition, we obtain a rank metric convolutional code, whose associated distance d_R has been called sum rank distance, see [NPRV17]. If we set $n_j = m_1 = 1$ for $j = 1,\dots,t$, we get a convolutional code in the Hamming metric d_H. As (sum rank metric) convolutional codes are $\mathbb{F}_q[D]^k$-submodules of $\mathbb{F}_q[D]^n$, in the following, we introduce some properties of polynomial matrices and polynomial modules.

Definition 3. *Let $G(D) = \sum_{i=0}^{\mu} G_i D^i \in \mathbb{F}_q[D]^{k\times n}$ with $G_\mu \neq 0$ and $k \leq n$. For each $i = 1,\dots,k$, the i-th **row degree** ν_i of $G(D)$ is the largest degree of any entry in row i of $G(D)$. The **external degree** of $G(D)$ is the sum of its row degrees. The **internal degree** of $G(D)$ is the maximal degree of its $k\times k$ minors. If its external and internal degree are equal, $G(D)$ is called a **minimal** generator matrix of the convolutional code it generates. The **degree** δ of a convolutional code $\mathcal{C}$ is defined as the external degree of a minimal generator matrix of $\mathcal{C}$. We call such a code an (n,k,δ) convolutional code. A convolutional code $\mathcal{C} \subset R$ in the generalized sum rank-metric is denoted as $(n_1\times m_1,\dots,n_t\times m_t,k,\delta)$ convolutional code.*

Definition 4. *A convolutional code $\mathcal{C}$ is called **non-catastrophic** if any of its generator matrices $G(D)$ is left prime, i.e. if $\mathrm{rk}(G(\lambda)) = k$ for all $\lambda \in \overline{\mathbb{F}}_q$, where $\overline{\mathbb{F}}_q$ denotes the algebraic closure of $\mathbb{F}_q$.*

Note that non-catastrophic convolutional codes are **delay-free**, i.e. G_0 has full rank. A convolutional code admits a parity-check matrix if and only if it is non-catastrophic [VY97]. In the rest of this section, we introduce column distances in the Hamming metric and in the rank metric.

Definition 5. *For $j \in \mathbb{N}_0$, the **j-th column distance** of a convolutional code $\mathcal{C}$ (in the Hamming metric) is defined as*

$$d_j^c(\mathcal{C}) := \min\left\{\mathrm{wt}_H(v_0,\dots,v_j) \mid v(D) = v_0 + v_1 D + \cdots \in \mathcal{C} \text{ and } v_0 \neq 0\right\}.$$

Theorem 2. *[GLRS06] Let $\mathcal{C}$ be an (n,k,δ) delay-free convolutional code. For $j \in \mathbb{N}_0$, $d_j^c(\mathcal{C}) \leq (n-k)(j+1)+1$. Moreover, if $d_j^c(\mathcal{C}) = (n-k)(j+1)+1$ for some $j \in \mathbb{N}_0$, then $d_i^c(\mathcal{C}) = (n-k)(i+1)+1$, for $i \leq j$.*

Definition 6 ([NPRS17])**.** *For $j \in \mathbb{N}_0$, the* ***j-th column rank distance*** *of a $(n \times m, k, \delta)$ non-catastrophic rank metric convolutional code is*

$$d_j^{cr}(\mathcal{C}) = \min\{wt_R(V(D)_{|[0,j]}) \mid V(D) \in \mathcal{C} \text{ and } V_0 \neq 0\}$$

where $V(D) = \sum\limits_{i \in \mathbb{N}_0} V_i D^i$ and $V(D)_{|[0,j]} = \sum\limits_{i=0}^{j} V_i D^i$.

Theorem 3 ([NPRS17])**.** *Let $\mathcal{C}$ be an $(n \times m, k, \delta)$ non-catastrophic rank metric convolutional code. Then,*

$$d_j^{cr}(\mathcal{C}) \leq j\left(n - \left\lfloor \frac{k}{m} \right\rfloor\right) + n - \left\lfloor \frac{k-1}{m} \right\rfloor. \tag{1}$$

Theorem 4 ([NPRS17])**.** *Let $\mathcal{C}$ be an $(n \times m, k, \delta)$ non-catastrophic rank metric convolutional code. If $d_j^{cr}(\mathcal{C})$ reaches the upper bound (1), then $d_i^{cr}(\mathcal{C})$ reaches this bound for all $i \leq j$.*

Definition 7. *If the bound in (1) is achieved for $j = \mu$, then we call the code* ***Memory Maximum Sum Rank (m-MSR)****. If this bound is achieved for*

$$L = \left\lfloor \frac{n\left\lfloor \frac{\delta}{k} \right\rfloor + \left\lfloor \frac{\delta - k(\lfloor \frac{\delta}{k} \rfloor + 1)}{m} \right\rfloor + \lfloor \frac{k-1}{m} \rfloor + 1}{n - \lfloor \frac{k}{m} \rfloor} \right\rfloor,$$

we call the code ***Maximal Rank Profile (MRP)****. Note that it can be shown with simple calculations that $j = L$ is the largest integer for which the bound in (1) can be achieved. Note that this integer L has already been calculated in [NPRS17]. However, in that paper the formula contains a mistake which we corrected here.*

3 Generalized Singleton bound for sum rank metric convolutional codes

In this section, we present a Singleton-like bound for convolutional codes in the generalized sum rank metric, which implies the Singleton bound for rank metric convolutional codes and Hamming metric convolutional codes.

Theorem 5. *Let $\mathcal{C}$ be an $(n_1 \times m_1, \ldots, n_t \times m_t, k, \delta)$ sum rank metric convolutional code. Let $\alpha \in \{1, \ldots, t\}$ be the maximum value such that*

$$\left(\left\lfloor \frac{\delta}{k} \right\rfloor + 1\right) k - \delta - 1 \geq n_1 m_1 + \cdots + n_{\alpha-1} m_{\alpha-1} \tag{2}$$

and $n = n_1 + \cdots + n_t$. Then,

$$d_{SR}(\mathcal{C}) \leq n\left\lfloor \frac{\delta}{k} \right\rfloor + \sum_{j=\alpha}^{t} n_j - \left\lfloor \frac{(\lfloor \frac{\delta}{k} \rfloor + 1)k - \delta - 1 - \sum_{j=1}^{\alpha-1} n_j m_j}{m_\alpha} \right\rfloor.$$

Proof. Let $G(D)$ be a minimal generator matrix of $\mathcal{C}$ with row degrees $\nu_0 \geq \cdots \geq \nu_{k-\ell-1} > \nu_{k-\ell} = \cdots = \nu_{k-1} = \nu_{min}$. Consider now $u(D) = u = (0, \dots, 0, u_{k-\ell}, \dots, u_{k-1}) \in \mathbb{F}_q^k \setminus \{0\}$. Then, $v(D) = u(D)G(D)$ has degree at most ν_{min}. As $v_0 = (v_{0,1}, \dots, v_{0,\sum_{j=1}^t n_j m_j}) = uG(0)$, we can select $u_{k-\ell}, \dots, u_{k-1}$ such that $v_{0,1} = \cdots = v_{0,\ell-1} = 0$. This means that for

$$V(D) = \psi(v(D)) = (V_1(D), \dots, V_t(D)) = \left(\sum_{i\in\mathbb{N}} V_{1,i} D^i, \dots, \sum_{i\in\mathbb{N}} V_{t,i} D^i \right),$$

the $\beta - 1$ matrices $V_{1,0}, \dots, V_{\beta-1,0}$ are zero where β is the maximum value such that $\ell - 1 \geq \sum_{j=1}^{\beta-1} n_j m_j$. Also, the first $\left\lfloor \frac{\ell-1-\sum_{j=1}^{\beta-1} n_j m_j}{m_\beta} \right\rfloor$ rows of $V_{\beta,0}$ are zero. So we get

$$rank(V_{i,0}) \leq \begin{cases} 0 & \text{for } i = 1, \dots, \beta - 1 \\ n_\beta - \left\lfloor \frac{\ell-1-\sum_{j=1}^{\beta-1} n_j m_j}{m_\beta} \right\rfloor & \text{for } i = \beta \\ n_i & \text{for } i = \beta + 1, \dots, t \end{cases}.$$

With this we can calculate

$$\begin{aligned} wt_{SR}(V(D)) &= \sum_{j=1}^{t} \sum_{i=0}^{\nu_{min}} rank(V_{j,i}) = \sum_{j=1}^{t} \sum_{i=1}^{\nu_{min}} rank(V_{j,i}) + \sum_{j=1}^{t} rank(V_{j,0}) \\ &\leq \nu_{min} \sum_{j=1}^{t} n_j + n_\beta - \left\lfloor \frac{\ell - 1 - \sum_{j=1}^{\beta-1} n_j m_j}{m_\beta} \right\rfloor + \sum_{j=\beta+1}^{t} n_j. \end{aligned}$$

This bound is maximal for $\nu_{min} = \lfloor \frac{\delta}{k} \rfloor$ and $\ell = (\lfloor \frac{\delta}{k} \rfloor + 1)k - \delta$ which implies $\alpha = \beta$ and

$$wt_{SR}(V(D)) \leq n \left\lfloor \frac{\delta}{k} \right\rfloor + \sum_{j=\alpha}^{t} n_j - \left\lfloor \frac{(\lfloor \frac{\delta}{k} \rfloor + 1)k - \delta - 1 - \sum_{j=1}^{\alpha-1} n_j m_j}{m_\alpha} \right\rfloor. \qquad \square$$

If a code attains the upper bound from the previous theorem we call it **Maximum Sum Rank Distance (MSRD)** convolutional code.

Remark 1. *A rank metric convolutional code is a sum rank metric convolutional code with $t = 1$. This implies $n_1 = n$, $m_1 = m$, and for α as in* (2)*, one always has $\alpha = 1$. With these values, Theorem 5 simplifies to*

$$d_R(\mathcal{C}) \leq n \left(\left\lfloor \frac{\delta}{k} \right\rfloor + 1 \right) - \left\lceil \frac{(\lfloor \frac{\delta}{k} \rfloor + 1)k - \delta}{m} \right\rceil + 1,$$

which is exactly the Singleton bound for rank metric convolutional codes from [NPRV17]. Convolutional codes that reach this upper bound are called MRD convolutional codes. In [NPRV16], it has been shown that such codes have generic row degrees, i.e. $k \lceil \frac{\delta}{k} \rceil - \delta$ row degrees are equal to $\lfloor \frac{\delta}{k} \rfloor$ and the other row degrees are equal to $\lceil \frac{\delta}{k} \rceil$.

Remark 2. *A Hamming metric convolutional code is a sum rank metric convolutional code with $n_i = m_i = 1$ for $i = 1, \dots, t$. This implies $t = n$ and $\alpha - 1 = \left(\lfloor \frac{\delta}{k} \rfloor + 1\right) k - \delta - 1$. With these values, Theorem 5 simplifies to*

$$d_H(\mathcal{C}) \leq n \left\lfloor \frac{\delta}{k} \right\rfloor + n - \alpha + 1 = (n-k)\left(\left\lfloor \frac{\delta}{k} \right\rfloor + 1\right) + \delta + 1,$$

which is exactly the generalized Singleton bound from [RS99].

4 Construction of MSRD Convolutional Codes

In this section, we will present constructions for convolutional codes that reach the upper bound of Theorem 5 for $\alpha = 1$. In the following theorem, we show how to obtain an MSRD convolutional code for general $t \in \mathbb{N}$ from several MRD convolutional codes.

Theorem 6. *For $j = 1, \dots, t$, let $G^j(D) \in \mathbb{F}_q[D]^{k \times n_j m_j}$ be minimal generator matrices of $(n_j \times m_j, k, \delta)$ MRD convolutional codes $\mathcal{C}_j$ with $d_R(\mathcal{C}_j) = n_j(\lfloor \frac{\delta}{k} \rfloor + 1)$ and with row degrees in non-increasing order. Let $\mathcal{C}$ be the sum rank metric convolutional code generated by $G(D) = \begin{pmatrix} G^1(D) & \dots & G^t(D) \end{pmatrix}$. If $\alpha = 1$, then $\mathcal{C}$ is an $(n_1 \times m_1, \dots, n_t \times m_t, k, \delta)$ MSRD code.*

Proof. As $G(D)$ is a minimal generator matrix with the same row degrees as the $G^j(D)$, the degree of $\mathcal{C}$ is δ. From Theorem 5 with $\alpha = 1$ we get

$$d_{SR}(\mathcal{C}) \leq n \left(\left\lfloor \frac{\delta}{k} \right\rfloor + 1\right) - \left\lfloor \frac{\left(\lfloor \frac{\delta}{k} \rfloor + 1\right) k - \delta - 1}{m_1} \right\rfloor. \tag{3}$$

Since $\mathcal{C}_1$ is MRD and has minimum distance $n_1\left(\lfloor \frac{\delta}{k} \rfloor + 1\right)$, we get

$$d_R(\mathcal{C}_1) = n_1 \left(\left\lfloor \frac{\delta}{k} \right\rfloor + 1\right) - \left\lfloor \frac{\left(\lfloor \frac{\delta}{k} \rfloor + 1\right) k - \delta - 1}{m_1} \right\rfloor = n_1 \left(\left\lfloor \frac{\delta}{k} \right\rfloor + 1\right)$$

and hence, $\left\lfloor \frac{(\lfloor \frac{\delta}{k} \rfloor + 1)k - \delta - 1}{m_1} \right\rfloor = 0$. Thus, the upper bound (3) simplifies to $d_{SR}(\mathcal{C}) \leq n\left(\lfloor \frac{\delta}{k} \rfloor + 1\right)$. We now show that $\mathcal{C}$ reaches this upper bound. For each $0 \neq u(D) \in \mathbb{F}_q[D]^k$, we obtain

$$V(D) = \psi(v(D)) = \begin{pmatrix} \psi(u(D)G^1(D)) & \dots & \psi(u(D)G^t(D)) \end{pmatrix}.$$

Since $\psi(u(D)G^j(D))$ is a nonzero codeword of $\mathcal{C}_j$, we have

$$wt_{SR}(V(D)) \geq \sum_{j=1}^{t} wt_R(\psi(u(D)G^j(D))) \geq \sum_{j=1}^{t} n_j \left(\left\lfloor \frac{\delta}{k} \right\rfloor + 1\right). \qquad \square$$

Combining the previous theorem with a construction for MRD convolutional codes from [NPRV17], we obtain the following construction for MSRD convolutional codes.

Theorem 7. *For an arbitrary finite field $\mathbb{F}_q$, consider $A_j \in \mathbb{F}_q^{m_j \times m_j}$ with an irreducible characteristic polynomial and any full row rank matrix $X_j \in \mathbb{F}_q^{n_j \times m_j}$ for all $j = 1, \dots, t$. Assume that $m_j \geq n_j$ and $m_j \geq \delta + k$ hold for all $j = 1, \dots, t$. For $0 \leq i \leq \lfloor \frac{\delta}{k} \rfloor$, let*

$$G_i = \begin{pmatrix} \psi^{-1}(X_1 A_1^{ki}, \dots, X_t A_t^{ki}) \\ \psi^{-1}(X_1 A_1^{ki+1}, \dots, X_t A_t^{ki+1}) \\ \vdots \\ \psi^{-1}(X_1 A_1^{ki+k-1}, \dots, X_t A_t^{ki+k-1}) \end{pmatrix},$$

$$G_{\lfloor \frac{\delta}{k} \rfloor + 1} = \begin{pmatrix} \psi^{-1}(X_1 A_1^{k\lfloor \frac{\delta}{k} \rfloor + k}, \dots, X_t A_t^{k\lfloor \frac{\delta}{k} \rfloor + k}) \\ \vdots \\ \psi^{-1}(X_1 A_1^{k+\delta-1}, \dots, X_t A_t^{k+\delta-1}) \\ 0_{(k\lceil \frac{\delta}{k} \rceil - \delta) \times n} \end{pmatrix}.$$

Then, the matrix $G(D) = \sum_{i=0}^{\lfloor \frac{\delta}{k} \rfloor + 1} G_i D^i \in \mathbb{F}_q[D]^{k \times \sum_{j=1}^{t} n_j m_j}$ is the generator matrix of an MSRD $(n_1 \times m_1, \dots n_t \times m_t, k, \delta)$ convolutional code.

Remark 3. *In [NPSV24], the authors modify the construction for MRD convolutional codes from [NPRV17] that we use in the previous theorem a little to enable larger values for δ. This allows to obtain a similar improvement in our construction for MSRD convolutional codes but for reasons of space we skip the corresponding details here.*

5 Generalized sum rank column distance

In this section, we generalize the notions of column distance and column rank distance for convolutional codes. However, before doing so, we present a new construction for convolutional codes that are m-MSR and MRP since there are not many constructions for convolutional codes with optimal rank column distance available. Moreover, the following theorem shows how to obtain m-MSR and MRP codes from MRD block codes, a strategy that will be used later in this section to obtain convolutional codes with optimal column rank distances from MSRD block codes.

Theorem 8. *Assume $\lfloor \frac{k}{m} \rfloor = \lfloor \frac{k-1}{m} \rfloor$. Let $\mathcal{D} \subseteq \mathbb{F}_q^{n \times m}$ be an MRD block code with rank distance $d_R(\mathcal{D}) = n - \lfloor \frac{k}{m} \rfloor$, i.e. $dim(\mathcal{D}) = m(n - d_R(\mathcal{D}) + 1) = m(\lfloor \frac{k}{m} \rfloor + 1)$, and we can choose a basis $\{A_1, \dots, A_{m(\lfloor \frac{k}{m} \rfloor + 1)}\}$ of $\mathcal{D}$. Set*

$$G_i = \begin{pmatrix} \psi^{-1}(A_{ik+1}) \\ \psi^{-1}(A_{ik+2}) \\ \vdots \\ \psi^{-1}(A_{(i+1)k}) \end{pmatrix} \quad for \quad i = 0, \ldots, j \leq \left\lfloor \frac{m(\lfloor \frac{k}{m} \rfloor + 1)}{k} \right\rfloor - 1,$$

and $G(D) = \sum_{i \in \mathbb{N}_0} G_i D^i$. *Let* $\mathcal{C}$ *be the* $(n \times m, k, kj)$ *rank metric convolutional code with generator matrix* $G(D)$. *Then,* $d_j^{cr}(\mathcal{C})$ *achieves the bound from Theorem 3. Moreover,* $\mathcal{C}$ *is an m-MSR and an MRP code.*

Proof. Let $u = (u_0, \ldots, u_j) \in \mathbb{F}_q^{k(j+1)}$, $u_i = (u_{i,1}, \ldots, u_{i,k})$ for $i = 0, \ldots, j$, $v_i = u_0 G_i + u_1 G_{i-1} + \ldots, u_i G_0, V_i = \psi(v_i) = \sum_{\ell=1}^{k} u_{0,\ell} A_{ik+\ell} + \cdots + \sum_{\ell=1}^{k} u_{i,\ell} A_\ell$. Hence, for $u_0 \neq 0$, each V_i is a non-trivial linear combination of $A_1, \ldots, A_{m(\lfloor \frac{k}{m} \rfloor + 1)}$, i.e. a codeword of $\mathcal{D}$. Since $d_R(\mathcal{D}) = n - \lfloor \frac{k}{m} \rfloor$, one has $rank(V_i) \geq n - \lfloor \frac{k}{m} \rfloor$ for $i = 0, \ldots, j$ and thus, $wt_R(V(D)_{|[0,j]}) = (j+1)(n - \lfloor \frac{k}{m} \rfloor)$. By calculations it follows that $\mathcal{C}$ m-MSR and MRP. □

Remark 4. *If* $\lfloor \frac{k}{m} \rfloor = \lfloor \frac{k-1}{m} \rfloor$ *is not true, then* $m \mid k$ *and one gets a construction for m-MSR convolutional codes (over large finite fields) from [MBK16].*

Definition 8. *The* ***j-th column sum rank distance*** *of a non-catastrophic* $(n_1 \times m_1, \ldots, n_t \times m_t, k, \delta)$ *sum rank metric convolutional code is*

$$d_j^{csr}(\mathcal{C}) = \min\{wt_{SR}(V(D)_{|[0,j]}) \mid V(D) \in \mathcal{C} \text{ and } V_0 \neq 0\}, \quad \text{for } j \in \mathbb{N}_0,$$

where $V(D) = \sum_{i \in \mathbb{N}_0} (V_{1,i}, \ldots, V_{t,i}) D^i$ *and* $V(D)_{|[0,j]} = \sum_{i=0}^{j} (V_{1,i}, \ldots, V_{t,i}) D^i$.

Theorem 9. *Let* $\mathcal{C}$ *be an* $(n_1 \times m_1, \ldots, n_t \times m_t, k, \delta)$ *non-catastrophic sum rank metric convolutional code. Let* $\beta_1, \beta_2 \in \{1, \ldots, t\}$ *be the maximal values such that* $k - 1 - \sum_{i=1}^{\beta_1 - 1} n_i m_i \geq 0$ *and* $k - \sum_{i=1}^{\beta_2 - 1} n_i m_i \geq 0$. *Then,*

$$d_j^{csr}(\mathcal{C}) \leq \sum_{i=\beta_1}^{t} n_i - \left\lfloor \frac{k-1-\sum_{i=1}^{\beta_1-1} n_i m_i}{m_{\beta_1}} \right\rfloor + j \left(\sum_{i=\beta_2}^{t} n_i - \left\lfloor \frac{k - \sum_{i=1}^{\beta_2-1} n_i m_i}{m_{\beta_2}} \right\rfloor \right). \quad (4)$$

Proof. Let $G(D) = \sum_{i \in \mathbb{N}_0} G_i D^i$ be a generator matrix of $\mathcal{C}$. As $G(D)$ is left prime, we can assume, without loss of generality, that the first k columns of G_0 are linearly independent. We proceed via induction with respect to j.

For $j = 0$, we choose $u_0 \in \mathbb{F}_q^k \setminus \{0\}$ such that $v_0 = u_0 G_0$ has the first $k-1$ entries equal to zero, so $wt_H(v_0) \leq \sum_{\ell=1}^{t} n_\ell m_\ell - k + 1$. Then, for $(V_{1,0}, \ldots, V_{t,0}) = \psi(v_0)$, we have that $V_{1,0}, \ldots, V_{\beta_1 - 1, 0}$ are zero and the first $\left\lfloor \frac{k-1-\sum_{i=1}^{\beta_1-1} n_i m_i}{m_{\beta_1}} \right\rfloor$ rows of $V_{\beta_1, 0}$ are equal to zero. So we get

$wt_{SR}(V_0) \leq \sum_{i=\beta_1}^{t} n_i - \left\lfloor \frac{k-1-\sum_{i=1}^{\beta_1-1} n_i m_i}{m_{\beta_1}} \right\rfloor$. Assume now that the upper bound holds for j. Let $u(D) \in \mathbb{F}_q[D]^k, v(D) = u(D)G(D)$, $V(D) = \psi(v(D)) = \sum_{i\in\mathbb{N}} (V_{i,1}, \ldots, V_{i,t})D^i \in \mathcal{C}$ be such that $wt_{SR}(V(D)_{|[0,j]}) = d_j^{csr}(\mathcal{C})$. Choose u_{j+1} so that $v_{j+1} = u_{j+1}G_0 + u_{j-1}G_1 + \cdots + u_0 G_{j+1}$ has the first k entries equal to zero. This means that the matrices $V_{j+1,1}, \ldots, V_{j+1,\beta_2-1}$ are zero and the first $\left\lfloor \frac{k-\sum_{i=1}^{\beta_2-1} n_i m_i}{m_{\beta_2}} \right\rfloor$ rows of V_{j+1,β_2} are equal to zero and we get

$$d_{j+1}^{csr}(\mathcal{C}) \leq wt_{SR}(V(D)_{|[0,j+1]}) = d_j^{csr}(\mathcal{C}) + \sum_{\ell=1}^{t} rank(V_{j+1,\ell})$$
$$\leq \sum_{i=\beta_1}^{t} n_i - \left\lfloor \frac{k-1-\sum_{i=1}^{\beta_1-1} n_i m_i}{m_{\beta_1}} \right\rfloor + (j+1)\left(\sum_{i=\beta_2}^{t} n_i - \left\lfloor \frac{k-\sum_{i=1}^{\beta_2-1} n_i m_i}{m_{\beta_2}} \right\rfloor\right).$$
$\square$

Remark 5. *The previous Theorem generalizes the bound of Theorem 3 as $t = 1$ implies $\beta_1 = \beta_2 = 1$, and the bound of Theorem 2 as $n_i = m_i = 1$ for $i = 1, \ldots, t$ implies $t = n$, $\beta_1 = k$, $\beta_2 = k + 1$.*

Theorem 2 and Theorem 4 can be easily generalized as follows.

Theorem 10. *Let $\mathcal{C}$ be an $(n_1 \times m_1, \ldots, n_t \times m_t, k, \delta)$ non-catastrophic sum rank metric convolutional code. If d_j^{csr} reaches the upper bound (4), then d_i^{csr} reaches this bound for all $i \leq j$.*

The following theorem can be shown in a similar way as Theorem 8.

Theorem 11. *Assume $\beta := \beta_1 = \beta_2$, $\left\lfloor \frac{k-1-\sum_{i=1}^{\beta-1} n_i m_i}{m_\beta} \right\rfloor = \left\lfloor \frac{k-\sum_{i=1}^{\beta-1} n_i m_i}{m_\beta} \right\rfloor$. Let $\mathcal{D} \subseteq F$ be an MSRD sum rank metric block code with distance $d_{SR}(\mathcal{D}) = \sum_{i=\beta}^{t} n_i - \left\lfloor \frac{k-\sum_{i=1}^{\beta-1} n_i m_i}{m_\beta} \right\rfloor$ and $dim(\mathcal{D}) = \sum_{i=1}^{\beta-1} m_i n_i + m_\beta \left(\left\lfloor \frac{k-\sum_{i=1}^{\beta-1} n_i m_i}{m_\beta} \right\rfloor + 1 \right)$. Choose a basis $\{(A_{1,1}, \ldots, A_{t,1}), \ldots, (A_{1,dim(\mathcal{D})}, \ldots, A_{t,dim(\mathcal{D})})\}$ of $\mathcal{D}$ and let*

$$G_i = \begin{pmatrix} \psi^{-1}(A_{1,ik+1}, \ldots, A_{t,ik+1}) \\ \vdots \\ \psi^{-1}(A_{1,(i+1)k}, \ldots, A_{t,(i+1)k}) \end{pmatrix} \quad \text{for } i = 0, \ldots, j \leq \left\lfloor \frac{dim(\mathcal{D})}{k} \right\rfloor - 1$$

and $G(D) = \sum_{i\in\mathbb{N}_0} G_i D^i$. Let $\mathcal{C}$ be the $(n_1 \times m_1, \ldots, n_t \times m_t, k, kj)$ sum rank metric convolutional code we obtain from $G(D)$. Then, $\mathcal{C}$ achieves the bound (4) for all $i = 0, \ldots, j$.

With the preceding theorem we obtain convolutional codes with optimal column sum rank distances from MSRD block codes over the same field; constructions for MSRD block codes can be found in e.g. [MP24, BGLR21].

Even if the constructions of optimal codes in the (generalized) sum rank metric presented in this paper have restricted code parameters, they all can be defined over arbitrary (small) finite fields.

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